

Question	Answer	Marks
4(a)(i)	spring constant	B1
4(a)(ii)	area represents the work done (to extend the wire)	B1
	work done (to extend the wire) is equal to elastic potential energy	B1
4(b)(i)	$x_G = 0.39 \text{ mm}$ and $x_H = 0.29 \text{ mm}$	A1
4(b)(ii)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ and $F = kx$ or $E = \text{area under graph}$	C1
	for G: $E = \frac{1}{2} \times 2.0 \times (0.39 \times 10^{-3})$ or $\frac{1}{2} \times 5.1 \times 10^3 \times (0.39 \times 10^{-3})^2$ for H: $E = \frac{1}{2} \times 2.0 \times (0.29 \times 10^{-3})$ or $\frac{1}{2} \times 6.9 \times 10^3 \times (0.29 \times 10^{-3})^2$ $E_P = 3.9 \times 10^{-4} + 2.9 \times 10^{-4}$ $= 6.8 \times 10^{-4} \text{ J}$	A1
4(b)(iii)	$E = FL / Ax$ or stress / strain = FL / Ax	C1
	$A_G / A_H = 1 \times (0.29 \times 10^{-3}) / [1.5 \times (0.39 \times 10^{-3})]$	C1
	ratio = 0.50	A1

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Question	Answer	Marks
5(c)(i)	cross labelled X drawn	D1